

25-11-2024 (M)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: July 2024 – November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 54	Class: LY- Honours	Semester: VII (SVU 2020)
Programme: Honours Programme in Data Science & Analytics		
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the department: Computer Engineering	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain the concept of bias and variance in Machine Learning.	5
ii)	In a churn prediction model, the confusion matrix for a test set of 500 customers is as follows: True Positive (TP) = 120, False Positive (FP) = 50, True Negative (TN) = 200, False Negative (FN) = 130. Calculate the accuracy, precision, recall, and F1-score of the model.	5
iii)	List the applications of Deep learning models across various different domains.	5
iv)	How does cross-validation help in model evaluation?	5
v)	In a CNN, if the input image is 32x32 pixels, the filter size is 5x5, and the stride is 1, what will be the size of the output feature map after applying the convolution operation? Assume no padding is used.	5
vi)	Consider the following objective function: $J(\theta) = (\theta^2 - 6\theta + 9)$, Where the initial value of $\theta_0 = 2$, and the learning rate $\alpha = 0.1$, Using the gradient descent update rule, update the value of θ after the first iteration.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Which activation function is generally preferred for the output layer in a multiclass classification problem, and why?	5
ii)	What is the primary difference between Stochastic Gradient Descent and Stochastic Gradient Descent with Momentum regarding their update rule and how do they affect parameter updates during training?	5
	OR	
Q2 A	Explain the differences between shallow and deep networks in neural networks. Why do deep networks generally perform better than shallow networks?	10
Q2 B	Solve any One	10
i)	What is a Convolutional Neural Network (CNN), and how is it different from a traditional fully connected neural network?	10
ii)	Describe the basic architecture of a CNN. What are the key layers in a typical CNN model, and what is the purpose of each?	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the role of the hidden state in an RNN. How does it help the network capture temporal dependencies in sequential data?	10
ii)	Describe the architecture and functionality of Long Short-Term Memory (LSTM) networks. How do LSTMs solve the vanishing gradient problem?	10
iii)	Consider a simple RNN with the following parameters: Input vector at time step $t = 1$, $x_1 = [1, 2]$ Weight matrix for the input W_{ih} : $\begin{bmatrix} 0.5 & -0.1 \\ 0.3 & 0.7 \end{bmatrix}$ Weight matrix for the hidden state W_{hh} : $\begin{bmatrix} 0.2 & -0.4 \\ 0.1 & 0.5 \end{bmatrix}$ Initial hidden state $h_0 = [0.1, -0.2]$ and Bias vector $b_h = [0.1, -0.1]$ Calculate the hidden state h_1 at time step $t = 1$.	10
Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is word embedding and why is it used in Natural Language Processing (NLP)?	10
ii)	Explain the components of the Seq2Seq architecture, including the encoder, decoder, and how they work together in generating an output sequence.	10
iii)	Explain the concept of text classification in deep learning. How do deep learning models, such as CNNs and RNNs, handle text classification tasks?	10

Que. No.	Question	Max. Marks
Q5	Write a short note on any Four	20
i)	Vector space model	5
ii)	Different types of RNN	5
iii)	TF – IDF (Term Frequency-Inverse Document Frequency)	5
iv)	Basic Pre-Processing steps in NLP	5
v)	Benefits of Dropout technique in deep learning	5
vi)	Role of Pooling layer in CNN	5